Project by using Verilog HDL

1.0 Traffic light controller without timer

```
// Traffic light controller code
module traffic_light_controller (
        input clk,
        input rst,
        output reg red,
      output reg yellow,
      output reg green
);

parameter RED    = 2'b00;
parameter GREEN  = 2'b01;
parameter YELLOW = 2'b10;

reg [1:0] state, next_state;

always @(posedge clk or posedge rst) begin
        if (rst)
                state <= RED;
        else
                state <= next_state;
end

always @(*) begin
```

```
        case (state)
                RED:    next_state = GREEN;
                GREEN:  next_state = YELLOW;
                YELLOW: next_state = RED;
                default: next_state = RED;
        endcase
end

always @(*) begin
         red = 0;
       yellow = 0;
       green = 0;

         case (state)
                RED:    red = 1;
            GREEN:  green = 1;
            YELLOW: yellow = 1;
        endcase
end
endmodule
```

Testbench

```
// testbench of traffic light controller
module tb_traffic_light();
reg clk;
reg rst;
```

```
wire red;
wire yellow;
wire green;

traffic_light_controller dut
(.clk(clk),.rst(rst),.red(red),.yellow(yellow),.green(green));

always #5 clk = ~clk;

initial begin
        $monitor("Time=%0t |RED=%b YELLOW=%b
GREEN=%b",$time,red,yellow,green);
        clk = 0;
        rst = 1;
        #10 rst = 0;

        #100 $finish();
end

initial begin
        $fsdbDumpfile("dump.fsdb");
        $fsdbDumpvars(0,tb_traffic_light);
end
endmodule
```

Simulation Output

Time=0 |RED=1 YELLOW=0 GREEN=0

Time=15 |RED=0 YELLOW=0 GREEN=1

Time=25 |RED=0 YELLOW=1 GREEN=0

Time=35 |RED=1 YELLOW=0 GREEN=0

Time=45 |RED=0 YELLOW=0 GREEN=1

Time=55 |RED=0 YELLOW=1 GREEN=0

Time=65 |RED=1 YELLOW=0 GREEN=0

Time=75 |RED=0 YELLOW=0 GREEN=1

Time=85 |RED=0 YELLOW=1 GREEN=0

Time=95 |RED=1 YELLOW=0 GREEN=0

Time=105 |RED=0 YELLOW=0 GREEN=1

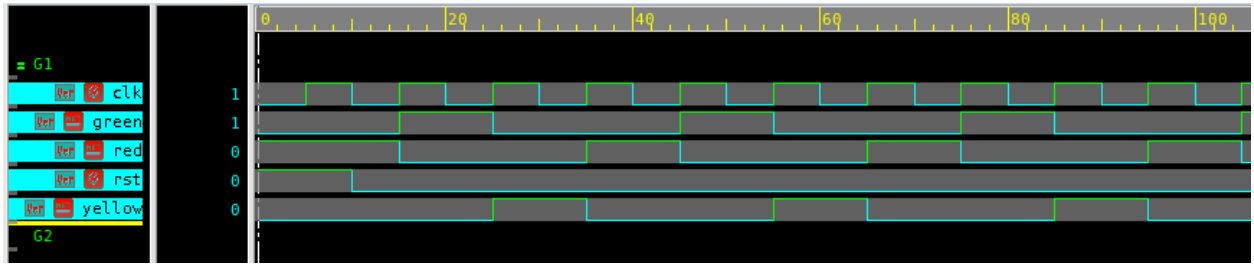
$finish called from file "traffic_code_tb.sv", line 19.

$finish at simulation time 110

V C S S i m u l a t i o n R e p o r t

Time: 110

Waveform



1.1 Traffic light controller code with timer

```
// Traffic light controller code with timer
module traffic_light_controller (
        input clk,
        input rst,
```

```
    output reg red,
    output reg yellow,
    output reg green
);

parameter RED    = 2'b00;
parameter GREEN  = 2'b01;
parameter YELLOW = 2'b10;

parameter RED_TIME    = 10;
parameter GREEN_TIME  = 10;
parameter YELLOW_TIME = 5;

reg [1:0] state, next_state;
reg [3:0] count;

always @(posedge clk or posedge rst) begin
    if (rst) begin
        state <= RED;
        count <= 0;
    end
    else begin
        if ((state == RED && count == RED_TIME) || (state == GREEN && count == GREEN_TIME) || (state == YELLOW && count == YELLOW_TIME)) begin
            state <= next_state;
```

```
                count <= 0;
            end

            else begin
                count <= count + 1;
            end
        end
end

always @(*) begin
        case (state)
            RED:    next_state = GREEN;
            GREEN:  next_state = YELLOW;
            YELLOW: next_state = RED;
            default: next_state = RED;
        endcase
end

always @(*) begin
         red = 0;
       yellow = 0;
       green = 0;

         case (state)
            RED:    red = 1;
```

```
            GREEN:  green = 1;
            YELLOW: yellow = 1;
        endcase
end
endmodule
```

Testbench

```
// testbench of traffic light controller
module tb_traffic_light();
reg clk;
reg rst;
wire red;
wire yellow;
wire green;

traffic_light_controller dut
(.clk(clk),.rst(rst),.red(red),.yellow(yellow),.green(green));

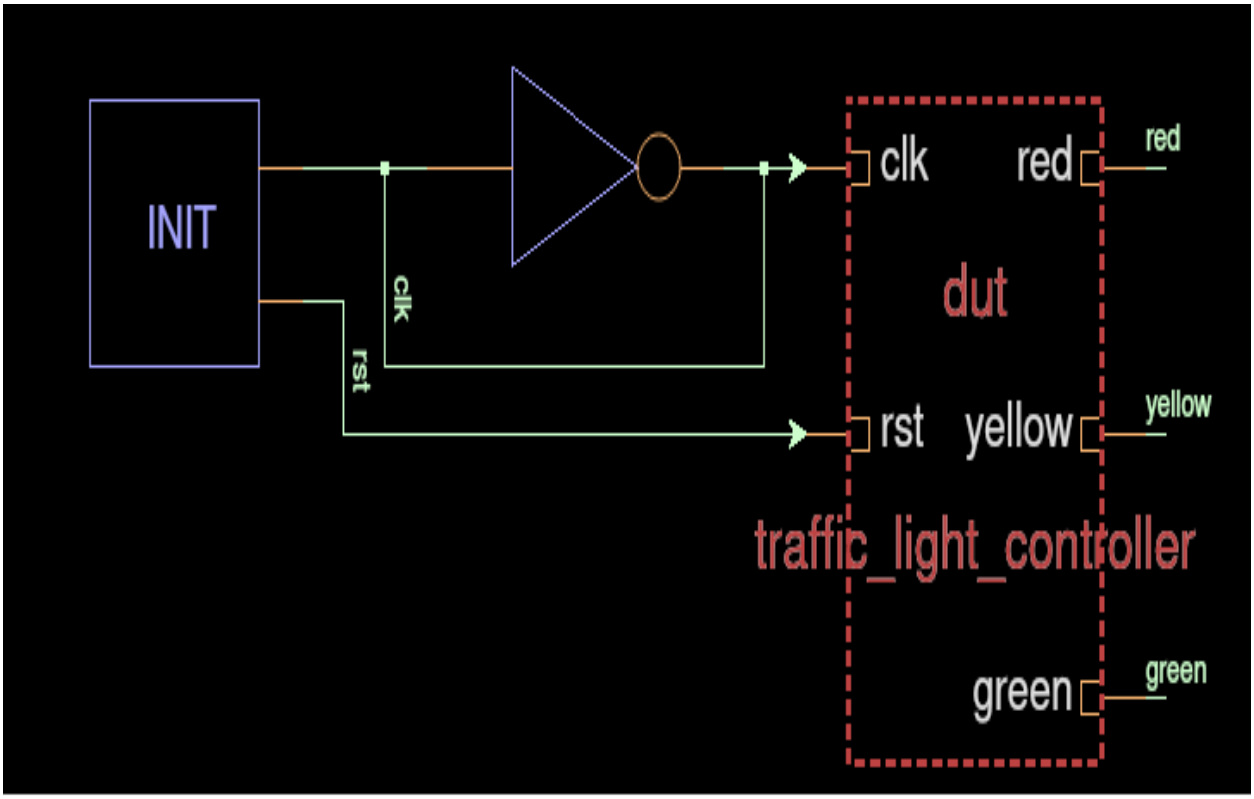
always #5 clk = ~clk;
initial begin
        $monitor("Time=%0t |RED=%b YELLOW=%b
GREEN=%b",$time,red,yellow,green);
        clk = 0;
        rst = 1;
        #10 rst = 0;

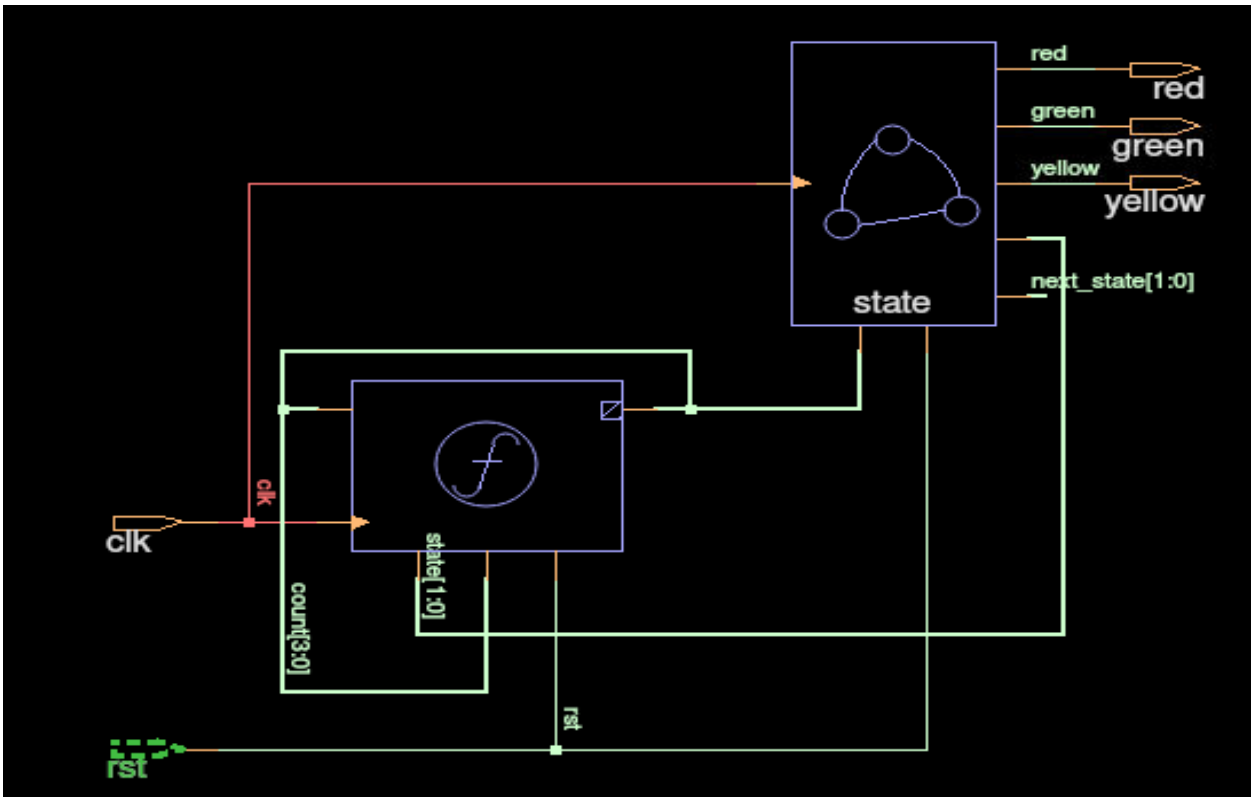
        #300 $finish();
```

```
end


initial begin
        $fsdbDumpfile("dump.fsdb");
        $fsdbDumpvars(0,tb_traffic_light);
end
endmodule
```

Simulation Results:





Time=0 |RED=1 YELLOW=0 GREEN=0

Time=115 |RED=0 YELLOW=0 GREEN=1

Time=225 |RED=0 YELLOW=1 GREEN=0

Time=285 |RED=1 YELLOW=0 GREEN=0

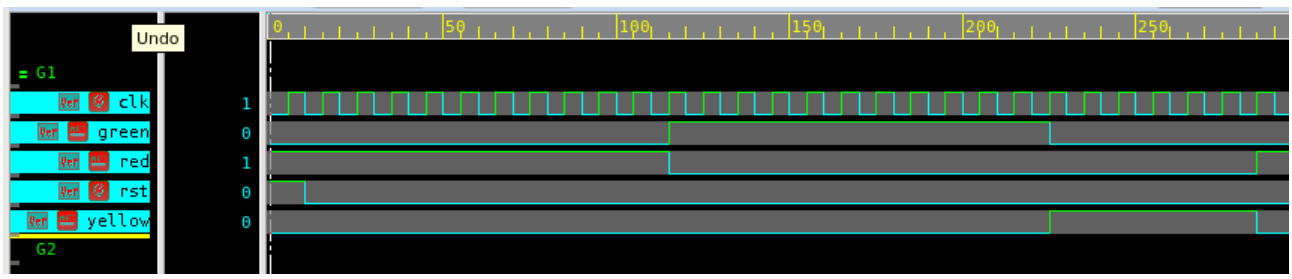
$finish called from file "traffic_code_tb.sv", line 19.

$finish at simulation time 310

V C S S i m u l a t i o n R e p o r t

Time: 310

Waveform



2.0 Vending Machine

```
// vending machine code
module vending_machine (
          input clk,
        input rst,
        input coin_5,
        input coin_10,
        output reg dispense,
        output reg change);

parameter S0  = 2'b00;
parameter S5  = 2'b01;
parameter S10 = 2'b10;

reg [1:0] state, next_state;

always @(posedge clk or posedge rst) begin
    if (rst)
        state <= S0;
    else
        state <= next_state;
end

always @(*) begin
    case (state)
```

```
        S0: begin
            if (coin_5)
                next_state = S5;
            else if (coin_10)
                next_state = S10;
            else
                next_state = S0;
        end

        S5: begin
            if (coin_5)
                next_state = S10;
            else if (coin_10)
                next_state = S10;
            else
                next_state = S5;
        end

        S10: begin
            next_state = S0;
        end

        default: next_state = S0;
    endcase
end
```

```
always @(*) begin
    dispense = 0;
    change = 0;

    case (state)
        S10: begin
            dispense = 1;
        end
    endcase

    if (state == S5 && coin_10)
        change = 1;
end
endmodule
```

Testbench

```
// testbench of vending machine
module tb_vending_machine;
reg clk;
reg rst;
reg coin_5;
reg coin_10;
wire dispense;
wire change;
```

```
vending_machine dut
(.clk(clk),.rst(rst),.coin_5(coin_5),.coin_10(coin_10),.dispense(dispense),.change(c
hange));


always #5 clk = ~clk;


initial begin
        $monitor("Time=%0t |coin_5=%b coin_10=%b |dispense=%b
change=%b",$time, coin_5,coin_10,dispense,change);


        clk = 0;
        rst = 1;
        coin_5 = 0;
        coin_10 = 0;


        #10 rst = 0;


        #10 coin_5 = 1; #10 coin_5 = 0;
        #10 coin_5 = 1; #10 coin_5 = 0;

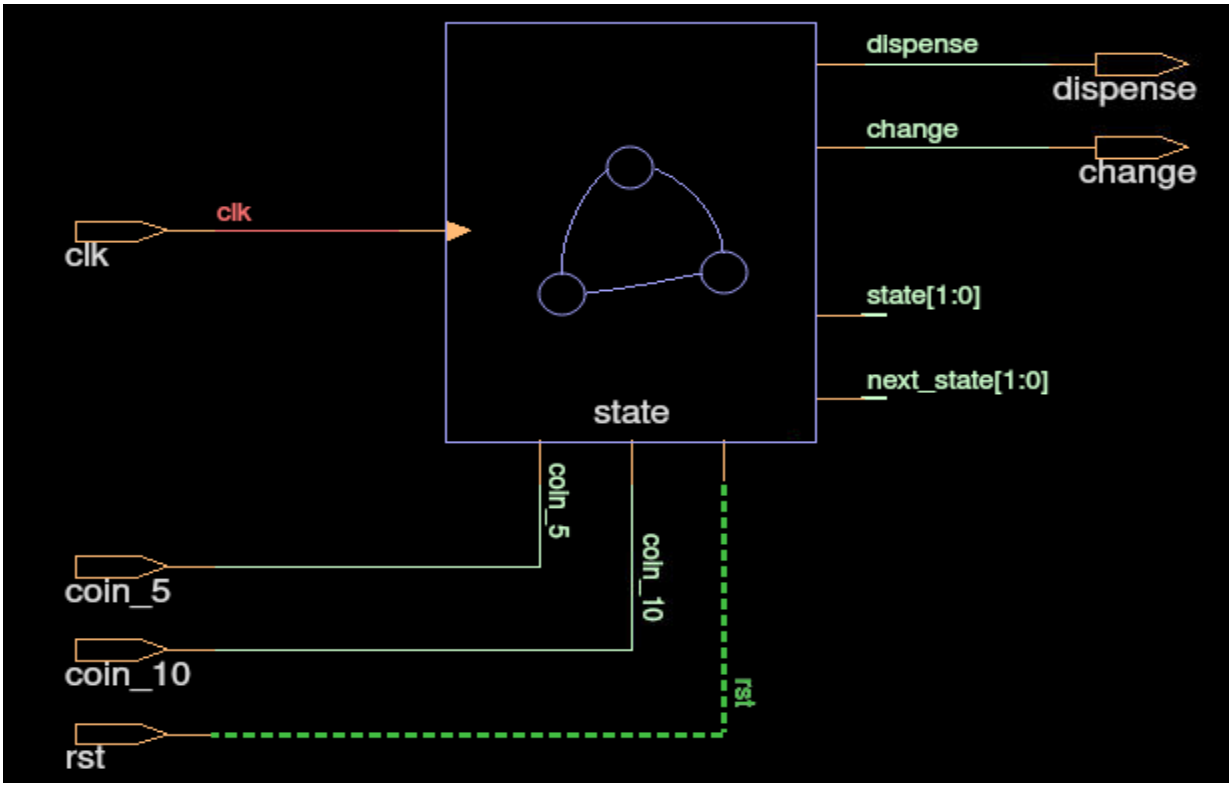

        #20 coin_10 = 1; #10 coin_10 = 0;

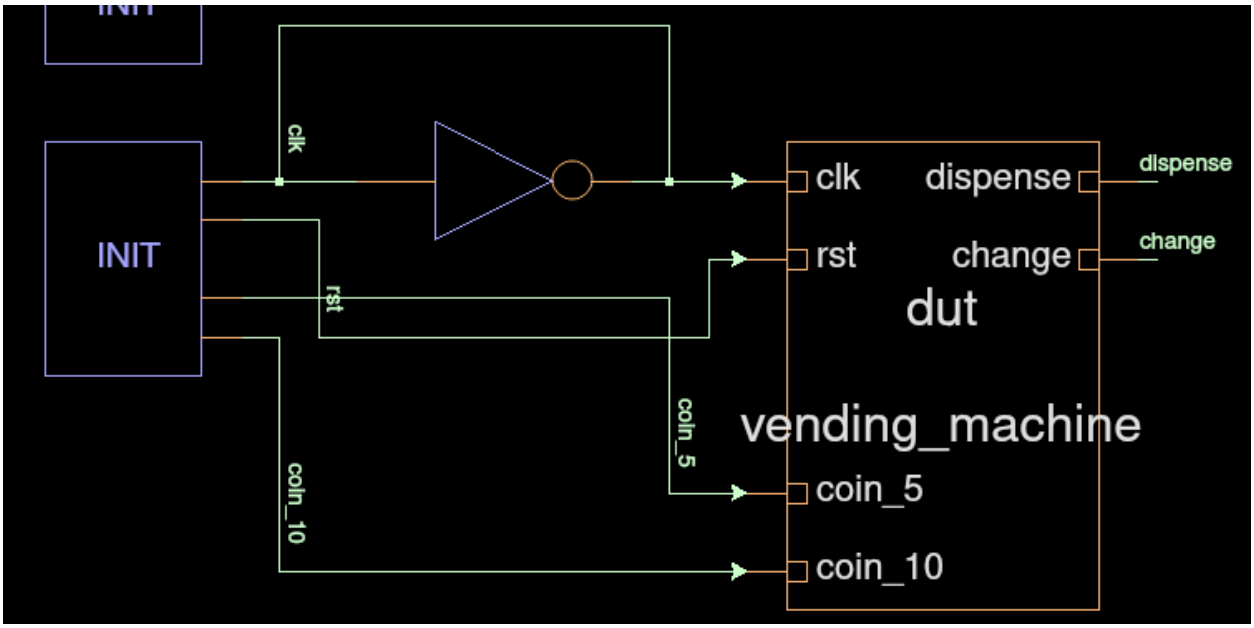

        #20 coin_5 = 1; #10 coin_5 = 0;
      #10 coin_10 = 1; #10 coin_10 = 0;


        #50 $finish;
```

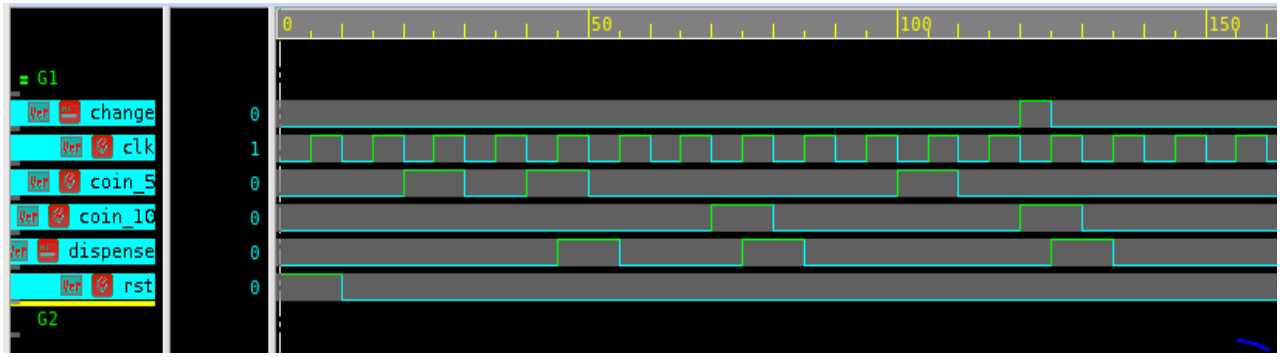
```
end


initial begin
    $fsdbDumpfile("dump.fsdb");
    $fsdbDumpvars(0, tb_vending_machine);
end
endmodule
```





Output waveform



Time=0 |coin_5=0 coin_10=0 |dispense=0 change=0

Time=20 |coin_5=1 coin_10=0 |dispense=0 change=0

Time=30 |coin_5=0 coin_10=0 |dispense=0 change=0

Time=40 |coin_5=1 coin_10=0 |dispense=0 change=0

Time=45 |coin_5=1 coin_10=0 |dispense=1 change=0

Time=50 |coin_5=0 coin_10=0 |dispense=1 change=0

Time=55 |coin_5=0 coin_10=0 |dispense=0 change=0

Time=70 |coin_5=0 coin_10=1 |dispense=0 change=0

Time=75 |coin_5=0 coin_10=1 |dispense=1 change=0

Time=80 |coin_5=0 coin_10=0 |dispense=1 change=0

Time=85 |coin_5=0 coin_10=0 |dispense=0 change=0

Time=100 |coin_5=1 coin_10=0 |dispense=0 change=0

Time=110 |coin_5=0 coin_10=0 |dispense=0 change=0

Time=120 |coin_5=0 coin_10=1 |dispense=0 change=1

Time=125 |coin_5=0 coin_10=1 |dispense=1 change=0

Time=130 |coin_5=0 coin_10=0 |dispense=1 change=0

Time=135 |coin_5=0 coin_10=0 |dispense=0 change=0

$finish called from file "vending_machine_tb.sv", line 32.

$finish at simulation time 180

V C S S i m u l a t i o n R e p o r t

Time: 180

3.0 Electronic Voting machine (EVM): -

```
// Electronic voting machine
module evm_3_candidate (
    input clk,
    input rst,
    input vote_en,
    input vote_a,
    input vote_b,
    input vote_c,

    output reg [7:0] count_a,
```

```
        output reg [7:0] count_b,
        output reg [7:0] count_c,
       output reg [7:0] total_votes);

always @(posedge clk or posedge rst) begin
   if (rst) begin
      count_a <= 0;
      count_b <= 0;
      count_c <= 0;
      total_votes <= 0;
   end
   else begin
      if (vote_en) begin

         if (vote_a && !vote_b && !vote_c) begin
            count_a <= count_a + 1;
            total_votes <= total_votes + 1;
         end

         else if (!vote_a && vote_b && !vote_c) begin
            count_b <= count_b + 1;
            total_votes <= total_votes + 1;
         end

         else if (!vote_a && !vote_b && vote_c) begin
```

```
            count_c <= count_c + 1;
            total_votes <= total_votes + 1;
        end

    end
  end
end
endmodule
```

Testbench

```
// Testbench of EVM
module tb_evm_3_candidate;
reg clk;
reg rst;
reg vote_en;
reg vote_a;
reg vote_b;
reg vote_c;

wire [7:0] count_a;
wire [7:0] count_b;
wire [7:0] count_c;
wire [7:0] total_votes;

evm_3_candidate uut
(.clk(clk),.rst(rst),.vote_en(vote_en),.vote_a(vote_a),.vote_b(vote_b),.vote_c(vote_
c),.count_a(count_a),.count_b(count_b),.count_c(count_c),.total_votes(total_votes)
```

```
);

always #5 clk = ~clk;

initial begin
        $monitor("Time=%0t |en=%b A=%b B=%b C=%b |countA=%0d countB=%0d countC=%0d total=%0d",$time, vote_en, vote_a, vote_b, vote_c,count_a, count_b, count_c, total_votes);

    clk = 0;
    rst = 1;
    vote_en = 0;
    vote_a = 0;
    vote_b = 0;
    vote_c = 0;

    #10 rst = 0;

    #10 vote_en = 1; vote_a = 1;
    #10 vote_a = 0;

    #10 vote_b = 1;
    #10 vote_b = 0;

    #10 vote_c = 1;
    #10 vote_c = 0;
```

```
    #10 vote_a = 1; vote_b = 1;
    #10 vote_a = 0; vote_b = 0;

    #10 vote_b = 1; vote_c = 1;
    #10 vote_b = 0; vote_c = 0;

    #10 vote_en = 0;
    #10 vote_a = 1;
    #10 vote_a = 0;

    #10 vote_en = 1;

    #10 vote_a = 1; #10 vote_a = 0;

    #10 vote_b = 1; #10 vote_b = 0;

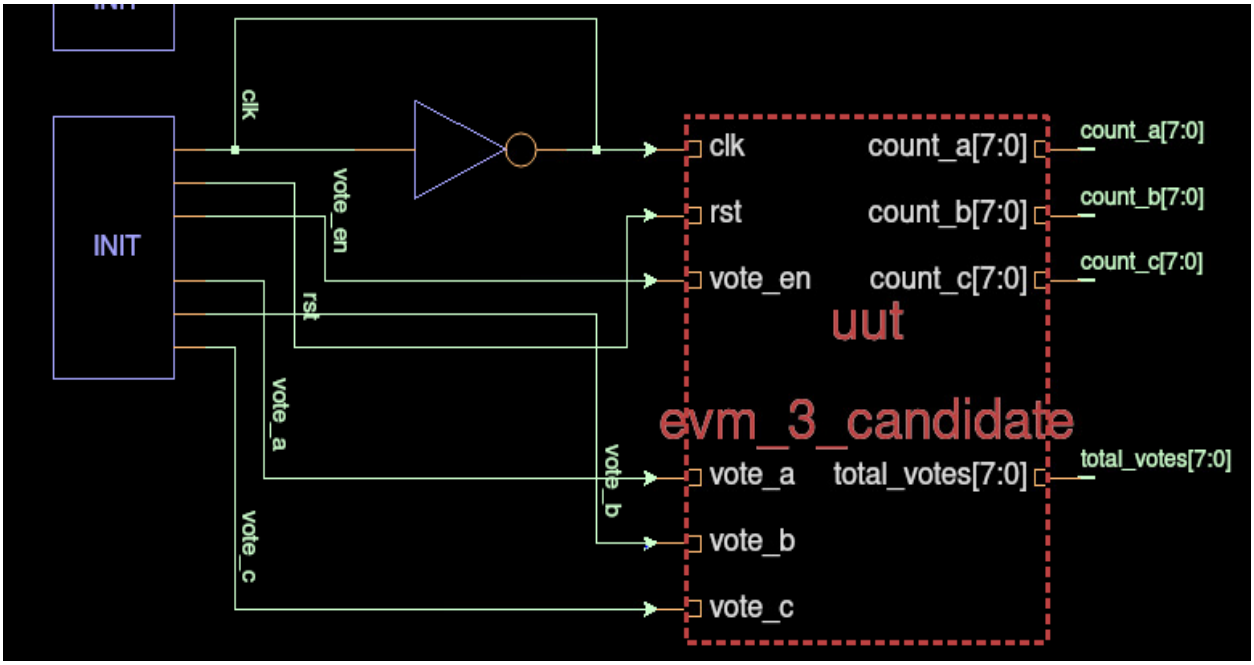
    #10 vote_c = 1; #10 vote_c = 0;

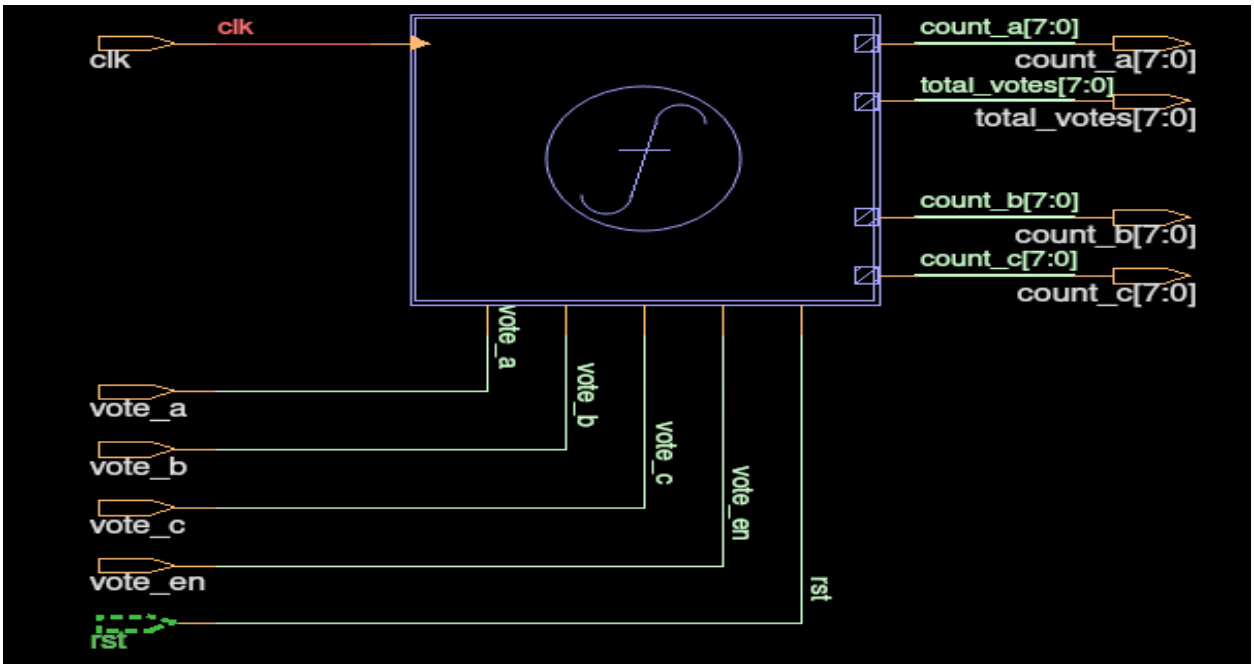
    #50 $finish;
end

initial begin
    $fsdbDumpfile("dump.fsdb");
    $fsdbDumpvars(0, tb_evm_3_candidate);
```

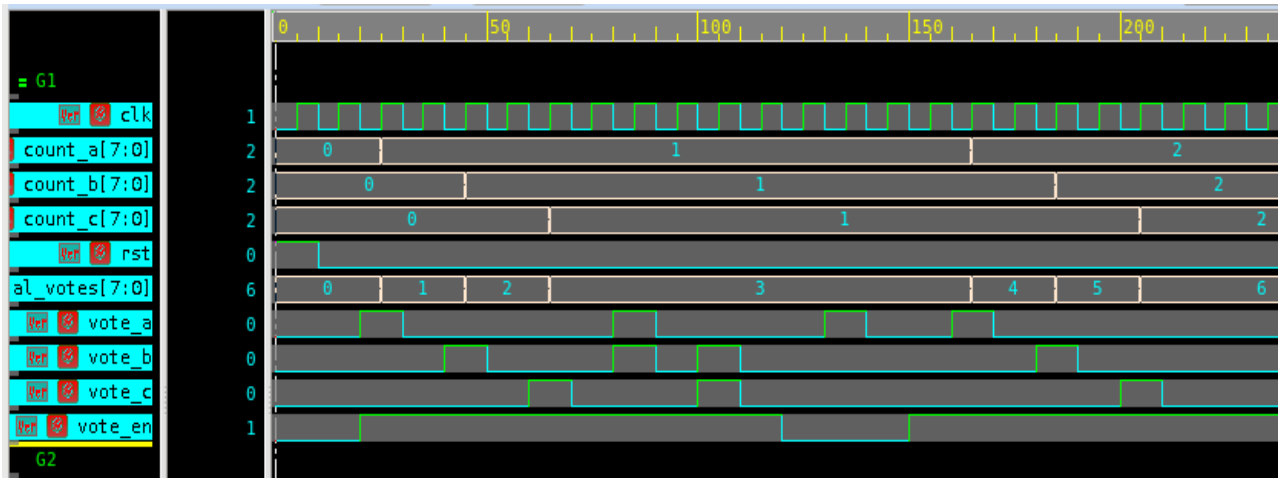
end

endmodule





Output waveform



Time=0 |en=0 A=0 B=0 C=0 |countA=0 countB=0 countC=0 total=0

Time=20 |en=1 A=1 B=0 C=0 |countA=0 countB=0 countC=0 total=0

Time=25 |en=1 A=1 B=0 C=0 |countA=1 countB=0 countC=0 total=1

Time=30 |en=1 A=0 B=0 C=0 |countA=1 countB=0 countC=0 total=1

Time=40 |en=1 A=0 B=1 C=0 |countA=1 countB=0 countC=0 total=1

Time=45 |en=1 A=0 B=1 C=0 |countA=1 countB=1 countC=0 total=2

Time=50 |en=1 A=0 B=0 C=0 |countA=1 countB=1 countC=0 total=2

Time=60 |en=1 A=0 B=0 C=1 |countA=1 countB=1 countC=0 total=2

Time=65 |en=1 A=0 B=0 C=1 |countA=1 countB=1 countC=1 total=3

Time=70 |en=1 A=0 B=0 C=0 |countA=1 countB=1 countC=1 total=3

Time=80 |en=1 A=1 B=1 C=0 |countA=1 countB=1 countC=1 total=3

Time=90 |en=1 A=0 B=0 C=0 |countA=1 countB=1 countC=1 total=3

Time=100 |en=1 A=0 B=1 C=1 |countA=1 countB=1 countC=1 total=3

Time=110 |en=1 A=0 B=0 C=0 |countA=1 countB=1 countC=1 total=3

Time=120 |en=0 A=0 B=0 C=0 |countA=1 countB=1 countC=1 total=3

Time=130 |en=0 A=1 B=0 C=0 |countA=1 countB=1 countC=1 total=3

Time=140 |en=0 A=0 B=0 C=0 |countA=1 countB=1 countC=1 total=3

Time=150 |en=1 A=0 B=0 C=0 |countA=1 countB=1 countC=1 total=3

Time=160 |en=1 A=1 B=0 C=0 |countA=1 countB=1 countC=1 total=3

Time=165 |en=1 A=1 B=0 C=0 |countA=2 countB=1 countC=1 total=4

Time=170 |en=1 A=0 B=0 C=0 |countA=2 countB=1 countC=1 total=4

Time=180 |en=1 A=0 B=1 C=0 |countA=2 countB=1 countC=1 total=4

Time=185 |en=1 A=0 B=1 C=0 |countA=2 countB=2 countC=1 total=5

Time=190 |en=1 A=0 B=0 C=0 |countA=2 countB=2 countC=1 total=5

Time=200 |en=1 A=0 B=0 C=1 |countA=2 countB=2 countC=1 total=5

Time=205 |en=1 A=0 B=0 C=1 |countA=2 countB=2 countC=2 total=6

Time=210 |en=1 A=0 B=0 C=0 |countA=2 countB=2 countC=2 total=6

$finish called from file "electronic_voting_machine_tb.sv", line 59.

$finish at simulation time 260

V C S S i m u l a t i o n R e p o r t

Time: 260